SOLUTIONS

Ballarat High School

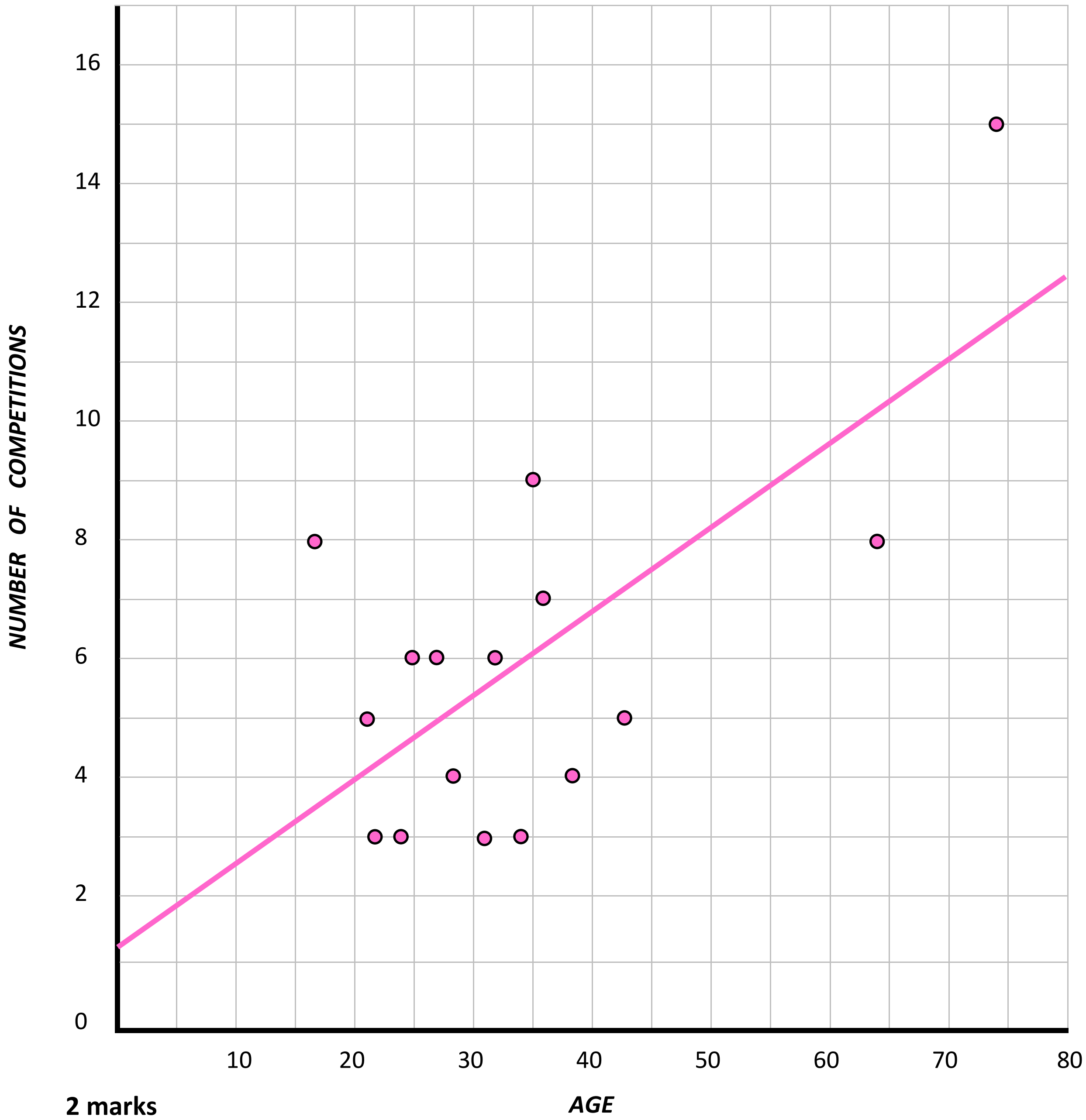
Data Analysis – Application Task (2022)

Part 2: The Competition

Section A

Section A

a)



2 marks

1 mark for scatterplot & 1 mark all data points correct place

b)

Age

1 for listing the explanatory variable

c)

Moderate Positive

1 mark for correct strength and direction – both must be correct.

d)

Number of comps competed in = 1.2 + 0.14 x Age

Number of comps = 1.2425 + 0.13634 x age

2 marks

1 mark for correct values to 2 significant figures

1 mark for correct variable names

e)

_________________*ANSWER ON GRAPH*________________

1 mark for line on graph. Must go through (0,1.2) and (80,12.4)

Will accept (0, 1-1.25) and (80, 12.1 – 12.4)

f)

The y intercept is 1.2. This means the number of comps competed in is 1.2 when the age is 0 years.

The slope is 0.14. This means the number of comps competed in increased by 0.14 for every 1 year increase in age.

4 marks

2 mark for y int interpretation – all units and variable names must be included correctly to get full 2 marks

2 mark for slope interpretation – all units and variable names must be included correctly to get full 2 marks

g)

Martha = 7 $1.2 + 0.14 \times 42 = 7.08$

Ricciardo = 3 $1.2 + 0.14 \times 15 = 3.3$

2 marks - 1 mark for each correct prediction to the nearest whole number

Incorrect rounding in prior question may impact unrounded predictions

h)

Martha is more reliable as this prediction is interpolation OR

Martha is more reliable as this involves predicting from within the data set

2 marks

1 mark for identifying Martha as the more reliable prediction

1 mark for reason

i)

Predicted value = $1.2 + 0.14 \times 24 = 4.56$

Residual = actual – predicted

Residual = $3 - 4.56$

Residual = -1.56

2 marks

1 mark for correct residual only to 2 decimal places

1 mark for full calculations

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Section B

Section B

a)

r= - 0.36

Negative because graph has a negative direction

1 mark for correct r value to 2 decimal places

b)

87%

1-0.1299 x 100 = 87.01

1 mark for correct percentage to the nearest whole number

c)

2nd round points = 45 – 1.2 x 1st round points

Find 2 points on the graph and input into biscatterplot – 4. Two points → equation

2nd round = 45-1.15 x 1st round

2 marks

1 mark for correct values to 2 significant figures

1 mark for correct variable names

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Section C

Section C

a)

BEST - The assumption of linearity is incorrect as the residual plot shows a definable hill shape

OR

ACCEPTABLE - The residual plot shows a hill shape this means the data is probably non-linear

2 marks

1 mark for comment of assumption of linearity

1 mark for describing residual plot

b)

log(x), $\frac{1}{x}$, y^2 OR

log(years of experience), $\frac{1}{years\ of\ experience}$,
$1st\ round\ points^2$

1 mark – must identify all three possible transformations (x,y or names of variables is acceptable)

c)

1^{st} round points^2 = 9.8 +7.8 x years of experience

2 marks

1 mark for correct values to 2 significant figures

1 mark for correct variable names

d)

15 points

$points^2 = 9.8 + 7.8 \times 26$

$points = 14.57$

1 mark for correct value to the nearest whole number

e)

Yes, predictions are more reliable

- As the residual plot is randomly scattered. This suggests the data has been linearised.
- as the coefficient of determination increased from 0.8836 to 0.9679.

1 mark for statement and evidence.

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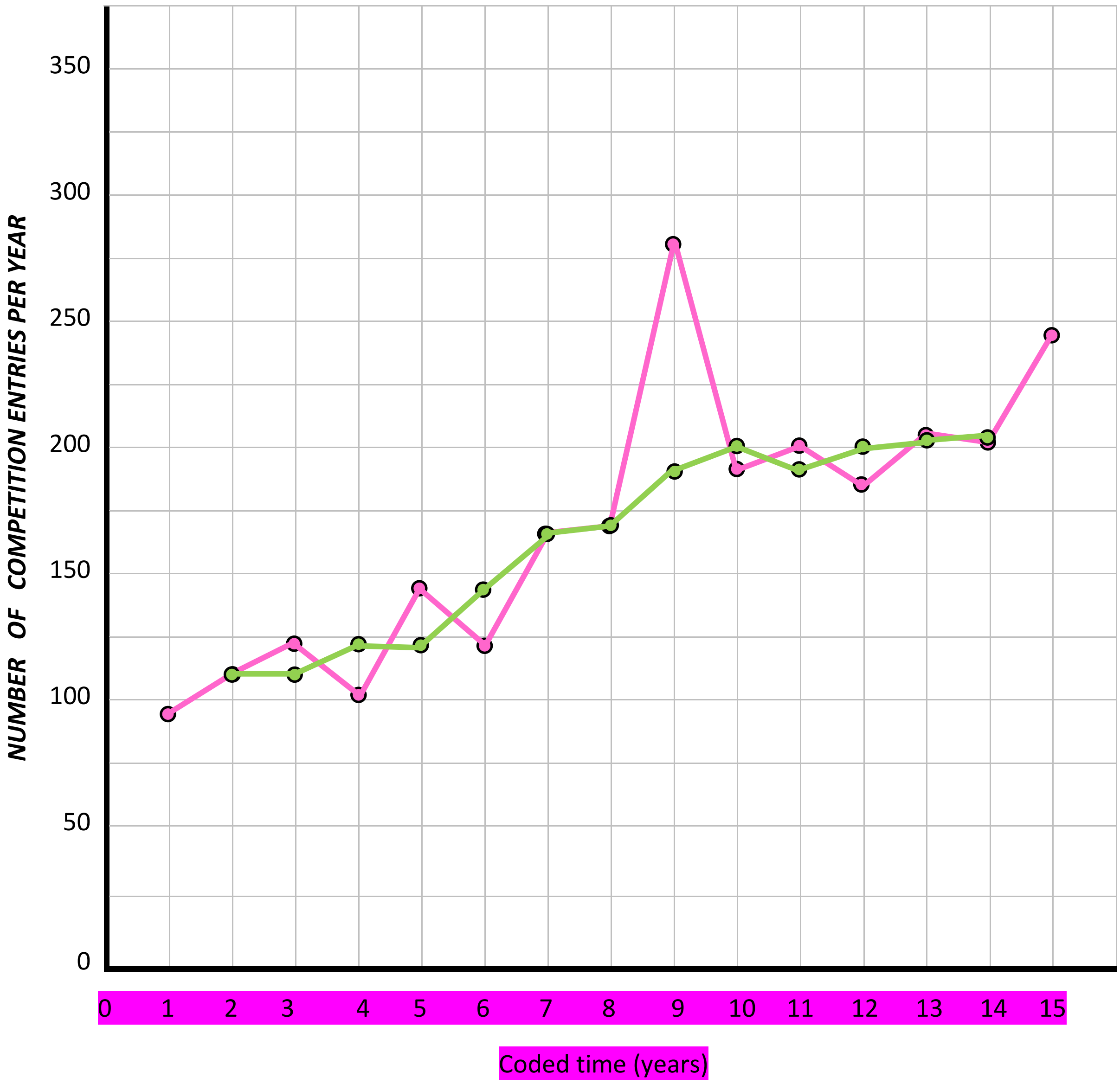
Section D

Section D

a)



3 marks

1 mark for label and axis of explanatory variable

1 mark for a time series plot & 1 mark for if all plot points are correct

b)

Overall increasing trend and an outlier at coded time 9

1 mark for both features

c)

__________________*ANSWER ON GRAPH*_________________

Green dots on graph.

2 marks if all correct

1 mark if one plot point is incorrect.

0 marks if more than one plot point is incorrect.

d)

$$\frac{143+122}{2}=132.5$$

$$\frac{122+167}{2}=144.5$$

$$\frac{132.5+144.5}{2}=138.5$$

1 mark for all lines of working out including the answer (do not round in working)

e)

Median smoothing as there is an outlier in the data OR

Median smoothing as this better smooths fluctuations.

1 mark for identifying smoothing technique and reason

f)

1.08

7-0.82-0.85-0.64-0.96-1.43-1.22

1 mark for correct seasonal index (do not round)

g)

The daily revenue for Sunday tends to be 22% higher than the daily average

1 mark for correct interpretation

h)

56%

$$\frac{0.36}{0.64} \times 100 = 56.25$$

1 mark for correct percentage to the nearest whole number

i)

$282

440 x 0.64 = 281.6

1 mark for correct reseasonalised value to the nearest dollar

END OF BOOKLET